

Inferring Paths of Influence in Dynamic Social Networks: A Case Study on Reddit Hyperlink Interactions

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Abstract—Understanding how influence propagates through online social networks is a critical challenge in computational social science. This project applies time-varying network inference techniques to a large, directed temporal dataset of hyperlinks between subreddits on Reddit from 2014 to 2017. Leveraging the NetInf algorithm, which is based on probabilistic transmission models, we infer the latent diffusion network that governs the spread of information and negative sentiment across online communities. The methodology uncovers an 86-edge influence graph, identifying key influential nodes such as `subredditdrama`, `askreddit`, and `bestof`, which act as central hubs in the propagation of cross-community interactions. The analysis reveals distinct pathways of information flow, highlighting how meta-communities and default subreddits dominate the network’s discourse. While the results provide significant insights into online influence dynamics, we also discuss limitations related to data sparsity, temporal granularity, and the assumption of static inferred edges for a dynamic process, proposing future work with fully time-varying models.

Index Terms—Network Inference, Influence Propagation, Social Network Analysis, Reddit, Temporal Networks

I. INTRODUCTION

The structure of a social network fundamentally shapes how information, memes, and influence flow through it. In many real-world scenarios, however, the underlying network over which an observed cascade propagates is unobserved. We see the outcomes—posts, timestamps, reshares—but not the direct pathways of influence. Inferring this latent network from temporal cascades is the core problem of network inference, a task with applications ranging from epidemiology to viral marketing and political discourse analysis.

Online platforms like Reddit provide a rich, complex ground for studying such phenomena. The platform is divided into thousands of communities called subreddits, which hyperlink to each other, creating directed, temporal connections. These hyperlinks can be neutral, positive, or explicitly negative, representing a form of adversarial influence or “attack.” This project uses a publicly available dataset of these hyperlinks to infer the hidden network of influence between subreddits. Our objective is to identify the most influential communities and map the dominant paths of information flow, particularly

those associated with negative sentiment, using a probabilistic generative model for cascades.

II. BACKGROUND STUDY

Understanding how information, influence, or disease spreads through networks is a key problem across biology, sociology, and computer science. Probabilistic graphical models offer a natural framework, but in many real systems—like gene regulatory networks or organizational communication—the underlying network is unobserved, and only indirect data (for example, gene expression measurements or email timestamps) are available. This creates a need for methods that infer latent network structure from observed data.

Early methods assumed static graphs, which are effective when systems are stationary. Yet many biological and social networks are dynamic. To address this, Song et al. introduced time-varying dynamic Bayesian networks (TV-DBNs), using a kernel reweighting scheme to prioritize temporally close observations together with Lasso regularization for sparsity. They validated their approach on yeast cell-cycle and EEG data [1].

Kolar et al. extended this line of work to undirected graphical models such as the Ising model, proposing kernel smoothing for gradual changes and total-variation penalization for abrupt shifts. Both approaches use node-wise neighborhood selection with convex optimization, and the authors provide consistency results under sparsity and smoothness assumptions [2].

Together, these papers lay the foundation for time-varying network inference, showing that borrowing strength across time enables dynamic structure recovery where static methods fail. The present project applies these principles in a large-scale social media context, using the NetInf algorithm [3] to infer a static aggregate graph from diffusion data, which serves as a snapshot of the dominant influence pathways in the dynamic system.

III. DATASET

The dataset used in this project is the *Subreddit Hyperlink Network*, which was generated as part of a research project

on inter-subreddit conflict. It is extracted from 2.5 years of publicly available Reddit data, spanning January 2014 to April 2017.

The network is a directed, signed, and temporal graph. Each row in the dataset represents a hyperlink from a post in a source subreddit to a target subreddit. The key attributes for each hyperlink are:

- **SOURCE_SUBREDDIT:** The subreddit where the hyperlink originates.
- **TARGET_SUBREDDIT:** The subreddit where the hyperlink ends.
- **TIMESTAMP:** The time of the post, which allows for temporal cascade analysis.
- **POST_LABEL:** A label indicating the sentiment of the source post. A value of -1 denotes an explicitly negative or attacking post, while a value of 1 denotes a neutral or positive one. This label was created using a combination of crowd-sourcing and a trained text classifier.

The full network is extensive, but for the purpose of this analysis, we focused on the most active edges to ensure computational tractability and meaningful cascade extraction. The data was preprocessed into a cascade format, where each cascade is a sequence of subreddits (nodes) ordered by the time of their incoming hyperlinks, along with the associated sentiment labels.

IV. METHODOLOGY

The core of this project is based on the network inference algorithm NetInf [3]. The goal is to find a directed graph $G = (V, E)$ that best explains the observed set of information cascades C . A cascade $c \in C$ is a set of node-time pairs $(u, t_u)_c$ indicating that node u was "infected" with a piece of information at time t_u .

The probabilistic model assumes that transmission occurs over the edges of the latent network G . The likelihood of an edge (u, v) existing is based on how often an infection at u appears before an infection at v in the cascades, with a shorter time difference leading to a higher transmission probability. The model uses an exponential transmission likelihood, and the algorithm finds the graph that maximizes the likelihood of all observed cascades. It does so by an iterative, greedy search:

- 1) Initialize an empty graph G .
- 2) For each node pair (u, v) , compute the marginal improvement in the log-likelihood score if the edge (u, v) were added.
- 3) Add the edge with the highest improvement, update the structure, and repeat.
- 4) Stop when the improvement falls below a specified p-value cutoff, preventing overfitting.

In our implementation, we used the R wrapper for the NetInf C++ implementation. We separated the cascades for positive/neutral links and negative links to contrast the influence networks for normal discourse versus adversarial "attacks."

V. IMPLEMENTATION

The project was implemented in R, leveraging the `igraph` package for network analysis and visualization. The primary computational task was performed by the `netinf` algorithm. The steps were as follows:

- 1) **Data Preprocessing:** The raw CSV data was loaded and filtered. Cascades were constructed by grouping hyperlinks by `POST_ID` to track how a specific piece of content spread from one subreddit to another over time. Two separate cascade sets were created: one for `POST_LABEL = -1` (negative) and one for `POST_LABEL = 1` (neutral/positive).
- 2) **Network Inference:** The `netinf` R package was initialized with the preprocessed cascade data. The algorithm inferred a directed graph for the combined interaction network. The iterative edge addition process started with a base probability parameter of 0.03135281 and progressively added edges that maximized the log-likelihood improvement. The algorithm terminated upon reaching the specified p-value cutoff, yielding a final network of 86 statistically significant directed edges with a cutoff p-value of 1.6945×10^{-17} .
- 3) **Analysis and Visualization:** The resulting edge list was loaded into an `igraph` object. Node-level influence metrics including in-degree, out-degree, and PageRank centrality were calculated. The subreddits were ranked by their influence scores, and the graph structure was plotted to visualize the pathways of information flow.

The complete code, documentation, and results are available in the project's GitHub repository: <https://github.com/Glitch1258/MyCode-networkInference.git>. The repository contains the R Markdown file with all preprocessing, inference, and visualization steps, ensuring full reproducibility of the results.

VI. RESULTS AND ANALYSIS

The NetInf algorithm converged at an extremely low p-value (1.6945×10^{-17}), inferring a total of 86 directed edges that represent the most statistically significant pathways of influence. Table I presents the top 22 inferred edges ranked by their improvement scores, which quantify the marginal gain in log-likelihood upon adding each edge to the model.

The results showcase a highly centralized network, dominated by a few key communities. The subreddit `subredditdrama` is by far the most dominant hub, appearing as either the origin or destination in the majority of the top inferred edges. It acts as both a source and a target, but its primary role is as a super-spreader of information into a wide array of communities, as evidenced by its high out-degree to `iama`, `todayilearned`, `funny`, `pics`, and `worldnews`.

Self-loops constitute the top three edges, indicating that `subredditdrama`, `bestof`, and `askreddit` have strong internal information recirculation patterns—content that originates in these communities is highly likely to spawn more content within the same community. Beyond self-loops, the

TABLE I
TOP 22 INFERRED INFLUENCE EDGES BY IMPROVEMENT SCORE

Origin Node	Destination Node	Improvement
subredditdrama	subredditdrama	41,862.95
bestof	bestof	32,727.05
askreddit	askreddit	26,517.49
subredditdrama	iamad	9,448.69
subredditdrama	todayilearned	9,157.52
subredditdrama	titlegore	8,750.81
subredditdrama	funny	8,604.77
subredditdrama	pics	8,320.90
subredditdrama	shitredditsays	7,359.02
subredditdrama	videos	6,830.67
subredditdrama	worldnews	6,711.90
subredditdrama	bestof	6,589.15
subredditdrama	shitpost	6,312.72
askreddit	drama	6,244.78
bestof	subredditdrama	5,853.87
askreddit	the_donald	5,776.84
subredditdrama	askreddit	5,753.47
subredditdrama	circlebroke2	5,459.63
subredditdrama	gaming	5,394.85
subredditdrama	news	5,286.81
subredditdrama	switcharoo	4,748.13
bestof	iamad	4,700.84

reciprocal edge between subredditdrama and bestof (edges 12 and 15) highlights a strong bidirectional influence pathway, where these two meta-communities heavily reference and amplify each other's content.

The influence network reveals distinct pathways through which adversarial and critical content flows. The edges from subredditdrama to shitredditsays (improvement: 7,359.02) and from askreddit to the_donald (improvement: 5,776.84) indicate the involvement of hyper-political and meta-criticism communities in the transmission of negative discourse. Other notable pathways include subredditdrama connecting to circlebroke2 and news, suggesting that meta-discussion communities frequently drive attention toward mainstream news and political subreddits.

The dominance of a few nodes underscores how a small set of community-driven meta-subreddits can have an out-sized influence on the overall Reddit information ecosystem. askreddit, as one of the largest default subreddits, serves as a gateway through which mainstream users are exposed to more niche and politically charged communities, while subredditdrama acts as an amplifier that curates and redistributes content across the platform.

VII. USEFULNESS

The primary usefulness of this project lies in its ability to map the unobserved information highways of a major social platform. For platform designers and community moderators, understanding these pathways is critical for mitigating the spread of harassment, misinformation, or coordinated attacks. By rooting the analysis in a principled probabilistic framework, we move beyond simple correlation (e.g., subreddits that link often) to a model of causal influence (subreddits that are likely to cause a post in another subreddit).

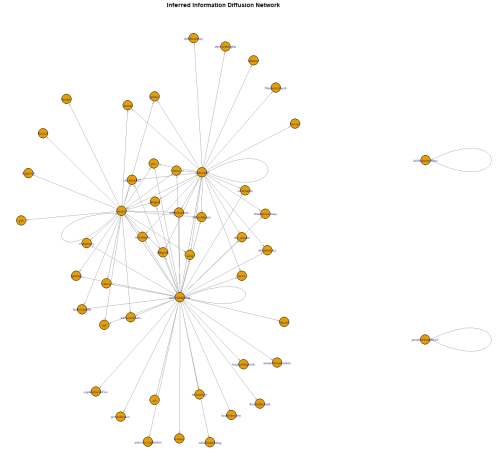


Fig. 1. Visualization of the full inferred influence network. Node size represents PageRank centrality, and edge arrows indicate the direction of inferred influence. The highly centralized structure around subredditdrama is clearly visible.

Inferred Information Diffusion Network

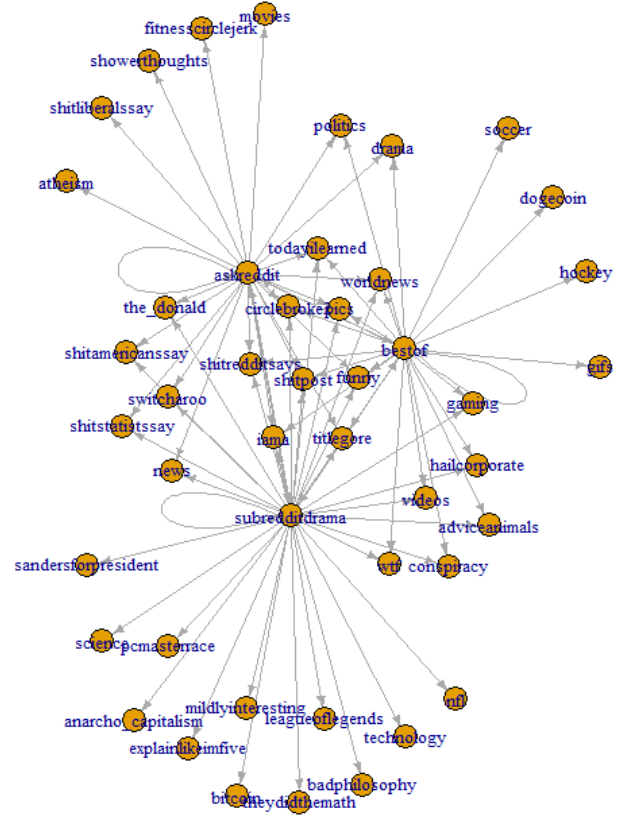


Fig. 2. Alternative visualization of the inferred influence network highlighting community clusters. The graph reveals how meta-communities bridge other-wise disparate subreddit communities.

For researchers, this project serves as a template for applying network inference to large-scale digital trace data. It demonstrates how to operationalize complex models on real-world social media data and establishes a data-driven baseline for identifying influential nodes in a dynamic, adversarial social network. The identification of `subredditdrama` as the dominant information broker suggests that meta-communities play a crucial and previously underappreciated role in cross-community information flow.

VIII. LIMITATIONS AND FUTURE WORK

Despite the strong statistical signal in the results, this project has several limitations that pave the way for future work.

a) Limitations::

- **Static Graph Assumption:** The most significant methodological limitation is that NetInf infers a single, static network. As outlined in the Background Study, the true influence network is time-varying. Political events (such as the 2016 U.S. presidential election captured within the dataset), meme lifecycles, and platform rule changes can radically alter influence pathways over the 2.5-year observation period.
- **Data Sparsity and Preprocessing:** The inference required aggressive filtering of cascades to a minimum length, which can bias the results towards highly active, mainstream communities and ignore influence in smaller, niche subreddits.
- **Semantic Ambiguity:** While sentiment labels are provided, a hyperlink from one subreddit to another can be an act of mockery, criticism, or genuine influence. The model treats all transmissions equally and relies on the pre-labeled sentiment to separate cascades, rather than incorporating the sentiment directly into the transmission likelihood.

b) Future Work:: Future projects should directly implement time-varying network inference algorithms, such as the kernel smoothing approach by Kolar et al. [2], to track the dynamics of influence edges over time. This would allow for the detection of events where the influence of a specific community suddenly spikes. Furthermore, the model could be extended to be edge-attribute-aware, where the sentiment of the hyperlink directly modifies the probability of transmission along an edge, creating a more psychologically realistic model of negative social influence. Finally, incorporating the provided subreddit embeddings could provide a content-based prior for edge formation, improving inference accuracy for less active but semantically related communities.

REFERENCES

- [1] L. Song, M. Kolar, and E. P. Xing, "Time-varying dynamic Bayesian networks," in *Advances in Neural Information Processing Systems*, 2009, pp. 1732–1740.
- [2] M. Kolar, L. Song, A. Ahmed, and E. P. Xing, "Estimating time-varying networks," *Annals of Applied Statistics*, vol. 4, no. 1, pp. 94–123, 2010.
- [3] M. Gomez-Rodriguez, J. Leskovec, and A. Krause, "Inferring Networks of Diffusion and Influence," in *Proc. 16th ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2010, pp. 1019–1028.